Public Spending In New York Towns

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Question:

Does government spending within NY towns come more from local taxes or government aid? (as seen through 3 metrics)

Specifically measuring:

- General Government Spending
- Public Safety
- Social Services

Data

Data Source: Office of the New York State Comptroller, Local Government Bulk Data

- 2967 observations across 914 NY Towns
- 11 years worth of data (2013-2023)

Converted to panel by appending each year into one dataset

gen_gov

Total amount of expenditures for services provided by the governmental entity for the benefit of the public or governmental body as a whole.

Includes:

- Administration
- Zoning and planning
- Operations
- Judgments
- County distribution of sales tax
- Miscellaneous general government

public_safety

Total amount of expenditures for police, fire and other public safety services.

Includes:

- Public safety administration
- Police
- Fire protection
- Emergency response
- Correctional services
- Disaster response
- Homeland security and civil defense
- Miscellaneous public safety

social_services

Total amount of expenditures for providing public assistance for individuals/families economically unable to provide essential needs for themselves.

Includes:

- Social service administration
- Financial assistance
- Medicaid
- Non-Medicaid medical assistance
- Housing assistance, employment services
- Youth services
- Public facilities
- Miscellaneous social services

fed_aid

Total amount of revenues derived from the federal government

Includes:

- Assistance with education
- Public safety
- Health
- Transportation
- Social services
- Economic development
- Culture and recreation
- Community services
- General government
- Utilities
- Sanitation
- Miscellaneous federal Aid

state_aid

Total amount of revenues derived from State Aid.

Includes:

- Education
- Public safety
- Health
- Transportation
- Social services
- Economic development
- Culture and recreation
- Community services
- General government
- Utilities
- Sanitation
- Miscellaneous State Aid

local_taxes

Total amount of tax revenue within town.

Includes:

- Property Tax
- Non-Property Tax (city income tax)
- Other property Tax (gains from sale of tax-acquired property)
- Sales & Use Tax (Sales Tax)

Clustering Standard Errors

Looks at each individual entity's standard error instead of lumping them all together.

- Not clustering <u>underestimates</u> standard errors

Coefficients don't change when clustering, only standard errors

Clustering Standard Errors

Without Clustered SE:

gen_gov	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
fed_aid	23.20462	.5793314	40.05	0.000	22.06858	24.34065
state_aid	29.29737	1.538265	19.05	0.000	26.28092	32.31382
local_taxes	.2887931	.6551799	0.44	0.659	-.9959775	1.573564
calendar_year						
2014	3356002	6389486	0.53	0.599	-9173416	1.59e+07
2015	2297298	6454991	0.36	0.722	-1.04e+07	1.50e+07
2016	2782282	6718582	0.41	0.679	-1.04e+07	1.60e+07
2017	-358776.1	6925177	-0.05	0.959	-1.39e+07	1.32e+07
2018	-1498538	6379024	-0.23	0.814	-1.40e+07	1.10e+07
2019	-3999640	6650776	-0.60	0.548	-1.70e+07	9042153
2020	152241.4	6590537	0.02	0.982	-1.28e+07	1.31e+07
2021	-2617357	6219223	-0.42	0.674	-1.48e+07	9578186
2022	-8499640	6072269	-1.40	0.162	-2.04e+07	3407734
2023	-4637340	6275034	-0.74	0.460	-1.69e+07	7667645
_cons	-3.98e+07	7417739	-5.36	0.000	-5.43e+07	-2.52e+07
sigma_u	73734348					
sigma_e	67757411					
rho	.54216706	(fraction of variance due to u_i)				

With Clustered SE:

gen_gov	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
fed_aid	23.20462	3.085513	7.52	0.000	17.14326	29.26598
state_aid	29.29737	15.32198	1.91	0.056	-.8020249	59.39676
local_taxes	.2887931	3.30908	0.09	0.930	-6.211757	6.789343
calendar_year						
2014	3356002	2354863	1.43	0.155	-1270028	7982033
2015	2297298	2397275	0.96	0.338	-2412048	7006644
2016	2782282	3161876	0.88	0.379	-3429092	8993656
2017	-358776.1	4570717	-0.08	0.937	-9337761	8620208
2018	-1498538	6026232	-0.25	0.804	-1.33e+07	1.03e+07
2019	-3999640	7856794	-0.51	0.611	-1.94e+07	1.14e+07
2020	152241.4	7832468	0.02	0.984	-1.52e+07	1.55e+07
2021	-2617357	9441915	-0.28	0.782	-2.12e+07	1.59e+07
2022	-8499640	1.07e+07	-0.79	0.428	-2.96e+07	1.26e+07
2023	-4637340	9904329	-0.47	0.640	-2.41e+07	1.48e+07
_cons	-3.98e+07	3.51e+07	-1.13	0.257	-1.09e+08	2.91e+07
sigma_u	73734348					
sigma_e	67757411					
rho	.54216706	(fraction of variance due to u_i)				

** Note regression coefficients do not change when clustering standard errors, only standard errors (red) and corresponding T statistics (blue)*

Clustering Standard Errors

	(1) gen_gov	(2) gen_gov	(3) gen_gov
fed_aid	14.47***	23.20***	23.20***
	(30.91)	(40.05)	(7.52)
state_aid	32.54***	29.30***	29.30
	(26.38)	(19.05)	(1.91)
local_taxes	-1.986***	0.289	0.289
	(-16.61)	(0.44)	(0.09)
2013.calen~r		0	0
		(.)	(.)
2014.calen~r		3356002.0	3356002.0
		(0.53)	(1.43)
2015.calen~r		2297298.1	2297298.1
		(0.36)	(0.96)
2016.calen~r		2782282.1	2782282.1
		(0.41)	(0.88)
2017.calen~r		-358776.1	-358776.1
		(-0.05)	(-0.08)
2018.calen~r		-1498538.1	-1498538.1
		(-0.23)	(-0.25)
2019.calen~r		-3999640.2	-3999640.2
		(-0.60)	(-0.51)
2020.calen~r		152241.4	152241.4
		(0.02)	(0.02)
2021.calen~r		-2617356.6	-2617356.6
		(-0.42)	(-0.28)
2022.calen~r		-8499639.6	-8499639.6
		(-1.40)	(-0.79)
2023.calen~r		-4637339.9	-4637339.9
		(-0.74)	(-0.47)
_cons	-14448868.0***	-39788691.4***	-39788691.4
	(-9.34)	(-5.36)	(-1.13)
N	2967	2967	2967

t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001

(1) Random Effects

(2) FE (no cluster)

(3) FE (cluster)

** Note the change in statistical significance for local_taxes before clustering standard errors and after (highly significant to insignificant after cluster)*

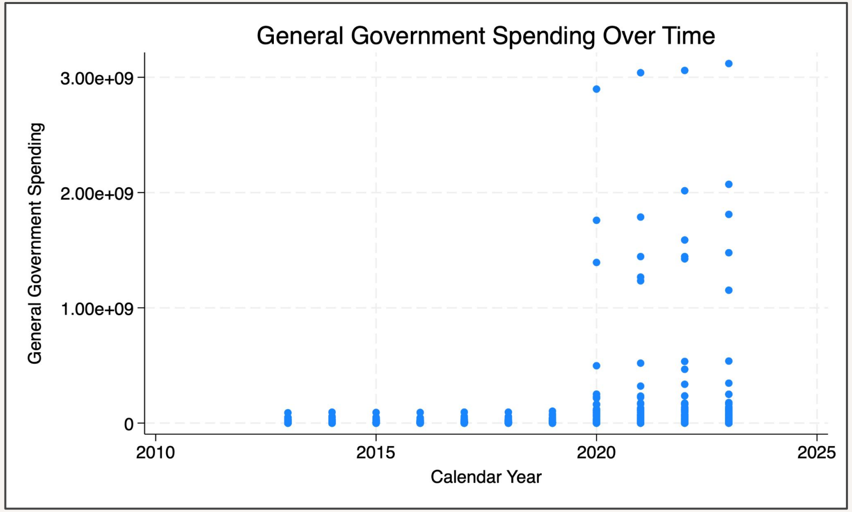
Clustering Standard Errors

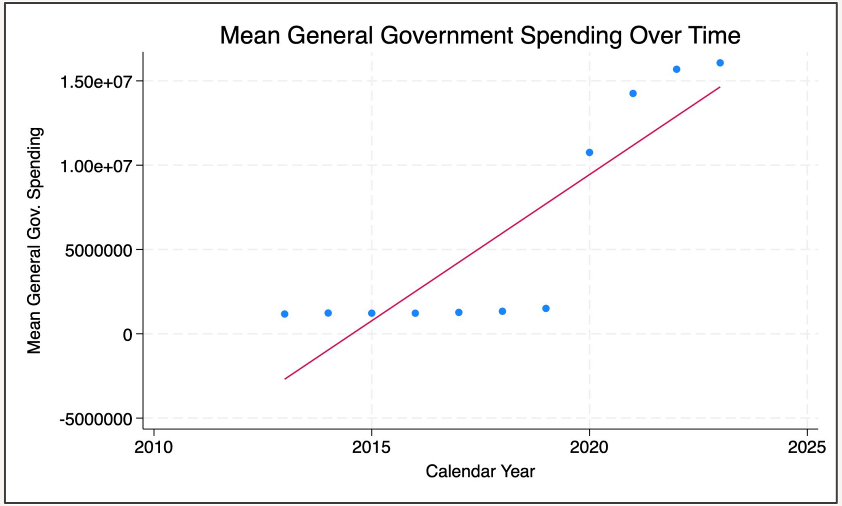
Potential Unit Fixed Effects:

- Sense of community within town
- Urban vs Rural status
- Vacation town status

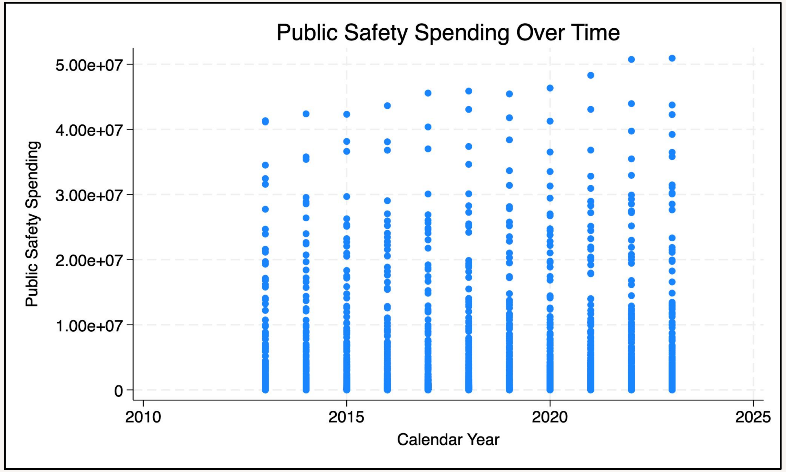
Potential Time Fixed Effects:

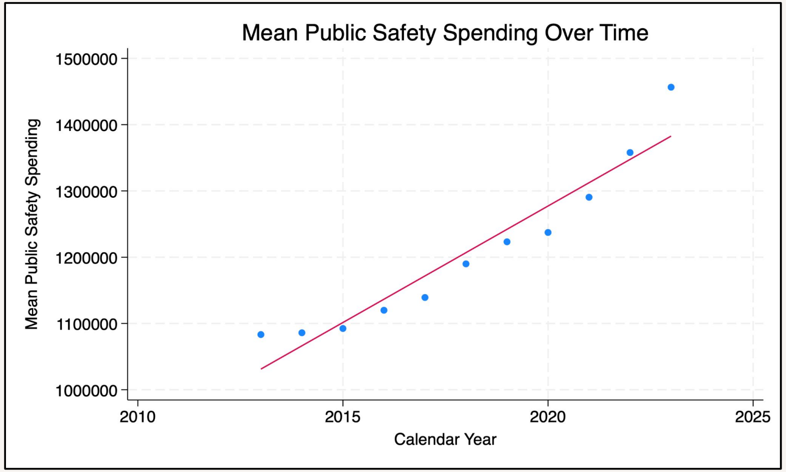
- COVID
- Political Regime changes
- Spending Contributions to Pensions (Large spike in 2019)

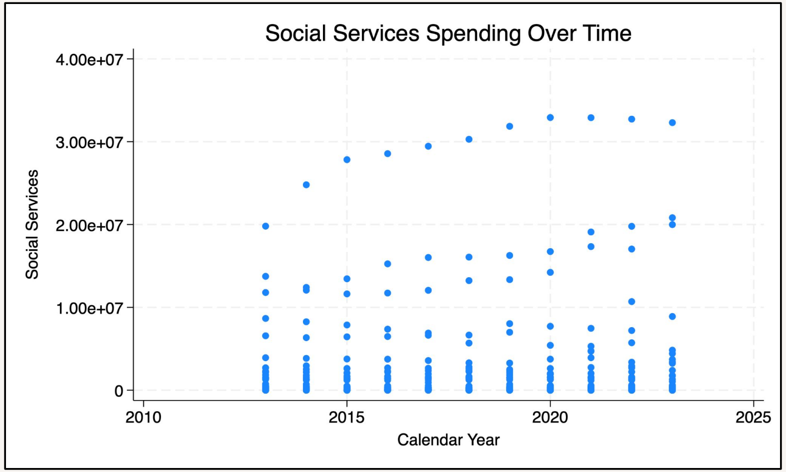


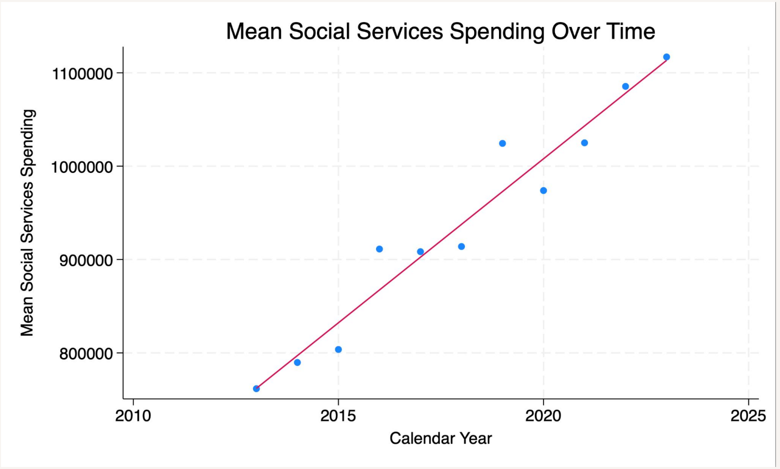


** Graphs are shown in dollars, not thousands*









** Graphs are shown in dollars, not thousands*

Findings

	(1) gen_gov	(2) public_saf~y	(3) social_ser~s
fed_aid	23.20***	0.0484***	0.00968
	(7.52)	(4.21)	(0.37)
state_aid	29.30	0.0338	-0.0946
	(1.91)	(0.59)	(-1.46)
local_taxes	0.289	0.174***	0.0387
	(0.09)	(9.52)	(0.88)
2013.calen~r	0	0	0
	(.)	(.)	(.)
2014.calen~r	3356002.0	-11858.9	-1362.2
	(1.43)	(-0.27)	(-0.02)
2015.calen~r	2297298.1	-2281.4	38444.1
	(0.96)	(-0.05)	(0.32)
2016.calen~r	2782282.1	20623.1	56390.0
	(0.88)	(0.36)	(0.40)
2017.calen~r	-358776.1	15922.0	57228.0
	(-0.08)	(0.33)	(0.37)
2018.calen~r	-1498538.1	45549.4	80978.3
	(-0.25)	(1.01)	(0.54)
2019.calen~r	-3999640.2	95038.1	109849.2
	(-0.51)	(1.65)	(0.68)
2020.calen~r	152241.4	23381.6	144007.8
	(0.02)	(0.48)	(1.01)
2021.calen~r	-2617356.6	71370.1	204280.8
	(-0.28)	(1.43)	(1.27)
2022.calen~r	-8499639.6	97063.8*	209250.9
	(-0.79)	(2.12)	(1.42)
2023.calen~r	-4637339.9	231862.8***	135966.5
	(-0.47)	(4.20)	(0.97)
_cons	-39788691.4	482746.7*	117296.8
	(-1.13)	(2.19)	(0.10)
N	2967	2967	604

t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001

** 2013 is reference year for LSDV Time Fixed Effects*

Findings

	(1) gen_gov	(2) public_saf~y	(3) social_ser~s
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2013.calen~r	0	0	0
	(.)	(.)	(.)
2014.calen~r	3356002.0	-11858.9	-1362.2
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2018.calen~r	-1498538.1	45549.4	80978.3
	(-0.25)	(1.01)	(0.54)
2019.calen~r	-3999640.2	95038.1	109849.2
	(-0.51)	(1.65)	(0.68)

General Government spending: mainly driven by state and federal aid

Public Safety spending: mainly driven by local taxes

Social Services spending: mainly driven by federal aid, but local taxes do contribute (Statistical insignificance)

Findings

Within towns in NY:

- Most of the general spending comes from government aid
- Local taxes are the main contributor to public safety spending
- Federal aid and local taxes contribute highest to social services spending (despite statistical insignificance)

Extensions:

- Expand the data outside of just NY, possibly Tri-state area or entire US
- Expand data for more years than just from 2013-2023
- Control for more variables (population, political regime, median income)

Thank you